



Certifications are important tools to show the prowess in the field by any person who holds them and gives recognition to stand out in the crowd"

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tems used for storing databases, diagnosing, performance optimisation, regular backups, database optimisation over mainframes, and ensuring database efficiency.

The people one works with are the management, IT professionals, end-users, system analysts, and technical support.

In any strategic decision relating to purchase of systems or equipment, or applying solutions, advice from these professionals is needed.

Data Security: Preparing data security plans, selection of appropriate tools, deployment of security tools, regular data backups, and selection of appropriate media for data is the responsibility of those in this segment.

Database Designers: Data helps in decision-making for business activities; for that, the system needs to be efficiently designed. A lot of data is accessed online in real-time, for, say, retailing. Data structures are to be designed such that there is no flaw at any end, and the structure has to be flexible enough to incorporate expansion.

Solution Architects: Those employed in this area interact with clients and derive solutions for their businesses. For example, in the case of the banking industry, solutions for real-time banking issues, data mobilisation and safety would be provided by the solution architects. They gather the client's requirements, derive appropriate solutions, document the solutions employed, and ensure solution integrity.

Data Warehousing, Business Intelligence, and Data Mining: See box *Specialisations*.

Entry Into The Field

The industry seeks candidates with a fairly good academic record. Candidates should be from the IT or engineering backgrounds. According to industry guidelines, graduates (BCA/B.E./B.Tech.) and post graduates (MCA/M.E./M.Tech.) in the field of computer science or computer engineering remain the most sought-after. In certain cases, students of mathematics are preferred.

During the academic tenure of pursuing an IT or engineering degree, subjects relating to database management systems, data structures, and database design need to be studied. A selection of subjects that covers DBMS packages and programming languages has to be made by both IT and engineering students.

"Formal education is fairly important and essential; also, cases of non-IT people who enter and become successful are very rare. Technical education strengthens the basic fundamentals about the job," says Seema Ambastha, Director-Sales Consulting (Database Technologies), Oracle India Ltd, who offer solutions for businesses—databases, middleware, business intelligence, business applications, and collaboration.

One's institution must be recognised by the All India Council for Technical Education (AICTE). The AICTE is a statutory body that coordinates development of the technical education system across the country. It becomes an individual's responsibility to check for the recognition of the educational institutes with the AICTE before applying for admission. Also, the electives being offered should be confirmed beforehand.

"Certifications are positively recommended. Today, business clients appreciate people with certifications. Be it on the technical side or the business side, they get recognised for their abilities and stand out from the crowd. For those into business intelligence, business/commercial knowledge is necessary at any level of the hierarchy," says Sanjay Deshmukh, Business Development Director—India/SAARC, Business Objects, who specialise in business intelligence services.

India has a number of global DBMS software vendors who offer education and training for their software packages. The major ones are Oracle, IBM, Sybase, and Microsoft. The courses offered are from the entrant level to the specialist level, with specially-designed modules for each level. These professional certifications add value to your résumé.

For non-IT people, options are available in the allied fields of databases such as data mining and business intelligence. A post-graduate degree equivalent to Master of Business, or a commercial background with specialisation in a domain such as finance, or a Chartered Accountant degree, are required for analytical functions for positions such as Financial analyst and budgeting.

"There are exceptions in every field. Students from non-IT backgrounds can indeed enter this arena. For such profiles, good commercial or business knowledge is necessary for dealing with clients and organisations from different verticals," says Davey Brooks, managing partner—professional services, South-East Asia/India, Teradata (a division of NCR Corporation), which offers enterprise data warehousing and analytical solutions.

Skills

The basic skills required of a data professional would be expertise over any one area of database technology. The usual soft skills such as the ability to work in a team are also required. For the consultancy and business solutions functions, skills such as inter-personal and intra-personal skills are essential.

Testing Times

While hiring candidates, organisations look for three criteria—education, experience, and professional credentials. For those just starting off, fundamentals relating to database management systems and general aptitude are tested. At the time of hiring, aptitude tests that check the aspirant's fundamentals and basics relating to DBMS are conducted. Organisations consider the project(s) opted for during the academic tenure. Across industry verticals, organisations into development of DBMS software necessarily scrutinise academic records, projects undertaken, and DBMS fundamentals.

Training

Training sessions assume that freshers have hands-on experience with DBMS applications and platforms. However, software vendors and software development organisations have training modules designed according to the experience and interest of the employees.